

Correlated Stationarity for Two-Qubit Entanglement Robustness

A Two-Qubit Pointer-Selection Benchmark

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Abstract

Papers 51 and 58 tested pointer-state robustness in one-qubit Lindblad benchmarks. Paper 58 showed that stationarity balance is more predictive than raw population balance for open-system pointer stability. This paper extends that test to two qubits, where pointer robustness is no longer only a question of state survival but also of entanglement survival.

We simulate 1800 two-qubit Lindblad trajectories across product, Bell, partially entangled, and Haar-entangled initial states, and across six environment families: local dephasing, collective dephasing, pair-parity dephasing, amplitude damping, thermal mixing, and depolarizing noise. The target is an entanglement-robustness score combining final concurrence, concurrence retention, mutual-information retention, and final purity. All predictors are computed from the initial state and generator only.

The central comparison is between raw two-qubit population balance and correlated generator stationarity. Scalar correlated stationarity improves over scalar population balance:

$$R_{\text{pop}}^2 = 0.7232, \quad R_{\text{corr-stat}}^2 = 0.7831, \quad \Delta R^2 = +0.0599.$$

The full stationarity-sensitive defect-field model reaches $R^2 = 0.9160$ and Spearman correlation 0.9477 in ordinary cross-validation, strongly above the shuffled-defect null. However, the short-time generator baseline performs best under leave-channel-family-out transfer ($R^2 = 0.7800$), while the full defect-field model reaches $R^2 = 0.6033$.

The useful conclusion is therefore narrow: two-qubit entanglement robustness aligns better with correlated generator stationarity than with raw population balance, but robust cross-channel generalisation remains dominated by short-time open-system dynamics. This is a classical simulation benchmark, not a quantum-hardware experiment, a collapse model, or a derivation of quantum measurement.

1 Purpose

The one-qubit pointer benchmark of Paper 58 established a simple lesson: open-system balance should be defined relative to the generator. For a one-qubit Lindblad equation, a state can be a robust pointer state while being strongly population-imbalanced. Population equality is therefore not the right notion of balance for pointer robustness.

Two qubits add a new question:

Which entangled states survive the environment?

This is closer to decoherence theory than the one-qubit case, because different environments select different stable structures. Local dephasing tends to destroy Bell coherences. Collective dephasing can protect decoherence-free subspaces. Pair-parity dephasing can preserve parity-sector Bell coherences. Amplitude damping and depolarizing noise tend to suppress entanglement.

The purpose of this paper is not to replace decoherence theory. It is to test whether the MAAT stationarity lesson remains useful when the object to be protected is not a single-state pointer axis, but a correlated two-qubit structure.

2 Two-Qubit Lindblad Benchmark

The dynamics are generated by a standard two-qubit Lindblad equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (1)$$

The Hamiltonian includes local fields and an exchange coupling:

$$H = h_{x1}X_1 + h_{x2}X_2 + h_{z1}Z_1 + h_{z2}Z_2 + J(X_1X_2 + Y_1Y_2). \quad (2)$$

The sampled environment families are:

- local Z dephasing;
- collective Z dephasing;
- pair-parity ZZ dephasing;
- local amplitude damping;
- thermal mixed damping and excitation;
- depolarizing noise.

Initial states are sampled from:

- weakly perturbed product states;
- Bell- Φ states;
- Bell- Ψ states;
- partially entangled Φ -type states;
- partially entangled Ψ -type states;
- Haar-random entangled states.

All trajectories are integrated classically by a fourth-order Runge–Kutta method on the density matrix. No quantum device, cloud runtime, or experimental dataset is used.

3 Entanglement-Robustness Target

For a two-qubit density matrix ρ , concurrence is denoted by $\mathcal{C}(\rho)$. The benchmark target is

$$Y_{\text{ent}} = 0.58 \mathcal{C}(\rho_T) + 0.22 \frac{\mathcal{C}(\rho_T)}{\max(\mathcal{C}(\rho_0), 0.08)} + 0.10 I_{\text{ret}} + 0.10 \text{Tr}(\rho_T^2). \quad (3)$$

Here I_{ret} is the retained mutual information, clipped to the unit interval. This target is not a universal definition of entanglement robustness. It is a transparent benchmark target that rewards final entanglement, entanglement retention, retained correlation, and final purity. The weights are deliberately concentrated on final concurrence while still rewarding retention and purity, so the target remains an entanglement-survival score rather than a generic mixed-state quality score.

4 Population Balance and Correlated Stationarity

The population-balance coordinate is

$$B_{\text{pop}} = (1 - |\langle Z_1 \rangle|) (1 - |\langle Z_2 \rangle|). \quad (4)$$

This is high when both local reduced states are population-balanced in the computational basis. For entangled states this can be misleading: many Bell states have locally maximally mixed marginals even when their entanglement is fragile under a particular environment.

The correlated stationarity coordinate instead asks whether local and correlated observables are initially stable under the generator:

$$B_{\text{corr-stat}} = \frac{1}{1 + (0.65 D_{\text{local}} + 1.15 D_{\text{corr}} + 0.90 D_{\text{C}} + 0.25 D_{\text{pur}}) / 2.2}. \quad (5)$$

The components are:

$$D_{\text{local}} = |\text{Tr}(\mathcal{L}[\rho_0] Z_1)| + |\text{Tr}(\mathcal{L}[\rho_0] Z_2)|, \quad (6)$$

$$D_{\text{corr}} = \frac{1}{3} \sum_{O \in \{Z_1 Z_2, X_1 X_2, Y_1 Y_2\}} |\text{Tr}(\mathcal{L}[\rho_0] O)|, \quad (7)$$

$$D_{\text{C}} \approx \max \left(0, \frac{\mathcal{C}(\rho_0) - \mathcal{C}(\rho_0 + \epsilon \mathcal{L}[\rho_0])}{\epsilon} \right), \quad (8)$$

$$D_{\text{pur}} \approx \max \left(0, \frac{\text{Tr}(\rho_0^2) - \text{Tr}((\rho_0 + \epsilon \mathcal{L}[\rho_0])^2)}{\epsilon} \right). \quad (9)$$

Thus $B_{\text{corr-stat}}$ is high when the initial local probabilities, two-body correlations, concurrence, and purity are locally stable under the Lindblad generator.

5 Structural Features and Baselines

The scalar population model uses

$$(H_{\text{stat}}, B_{\text{pop}}, S_{\text{ent}}, V_{\text{corr}}, R_{\text{rob}}, F_{\text{scalar}}).$$

The scalar stationarity model replaces B_{pop} by $B_{\text{corr-stat}}$. The full defect-field model adds generator-local pressures: Lindblad residual norm, local drift, correlation drift, concurrence slope, purity decay, Bell-sector overlaps, decoherence-free-subspace overlaps, damping pressure, dephasing pressure, depolarizing pressure, and Hamiltonian coupling pressure.

The comparison baselines are:

- rates only;
- an einselection-inspired baseline using Bell-sector overlaps, decoherence-free-subspace overlaps, and channel pressures;
- a short-time generator baseline using initial concurrence, Lindblad norm, concurrence slope, purity decay, local drift, and correlation drift;
- shuffled defect-field nulls.

6 Results

6.1 Model Comparison

Feature set	RF R^2	RF Spearman	LFO R^2	LFO Spearman
Defect-field stationary	0.9160	0.9477	0.6033	0.6706
Short-time generator baseline	0.8060	0.8773	0.7800	0.7214
Scalar correlated stationarity	0.7831	0.8154	0.7286	0.5554
Einselection baseline	0.7574	0.8672	-0.0110	0.1659
Scalar population	0.7232	0.7899	0.6179	0.5055
Rates only	0.4871	0.8520	0.0357	0.1995
Shuffled defect null	-0.0493	-0.0048	-0.2176	-0.3966

Table 1: Two-qubit entanglement-robustness prediction. LFO denotes leave-channel-family-out transfer.

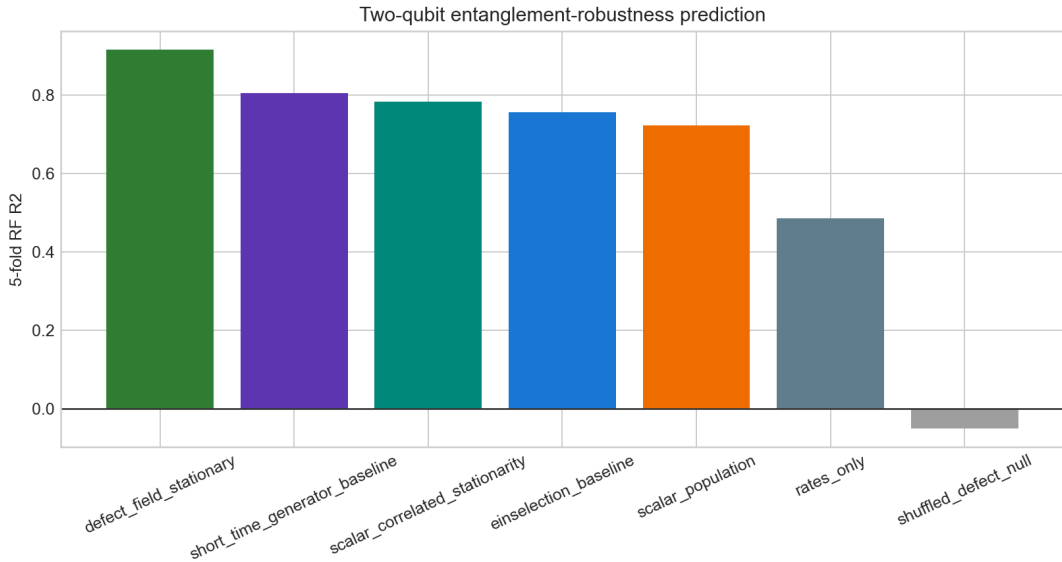


Figure 1: Ordinary cross-validation model comparison. The full stationarity-sensitive defect field performs best, while shuffled defects collapse.

The scalar comparison is the core controlled test:

$$\Delta R^2 = R_{\text{corr-stat}}^2 - R_{\text{pop}}^2 = 0.7831 - 0.7232 = 0.0599.$$

Thus correlated stationarity improves over raw population balance in this two-qubit benchmark.

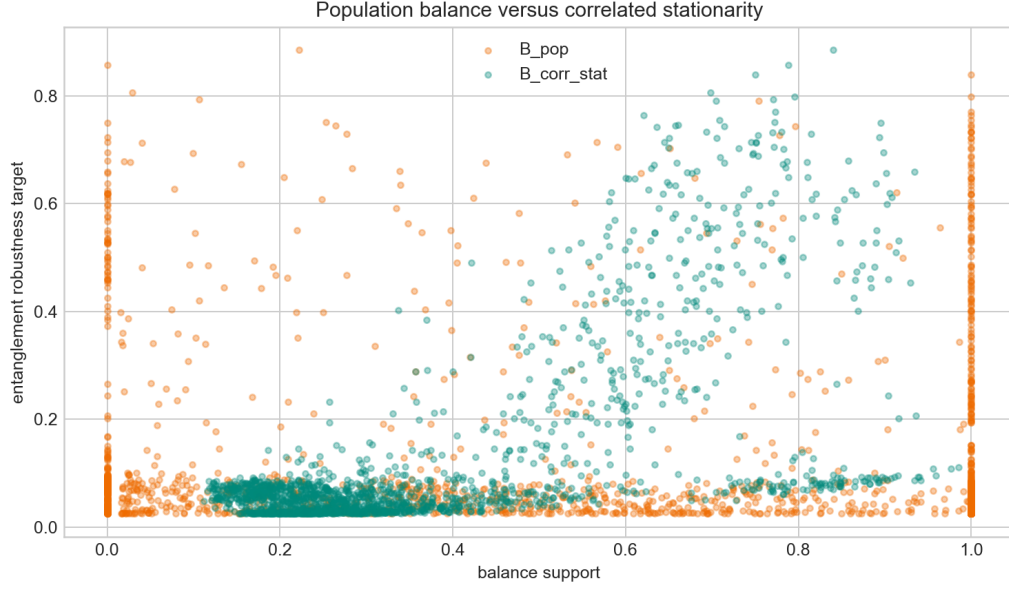


Figure 2: Population balance versus correlated stationarity. Raw local population balance does not distinguish robust from fragile Bell-like structures as cleanly as generator-sensitive correlated stationarity.

6.2 Channel Structure

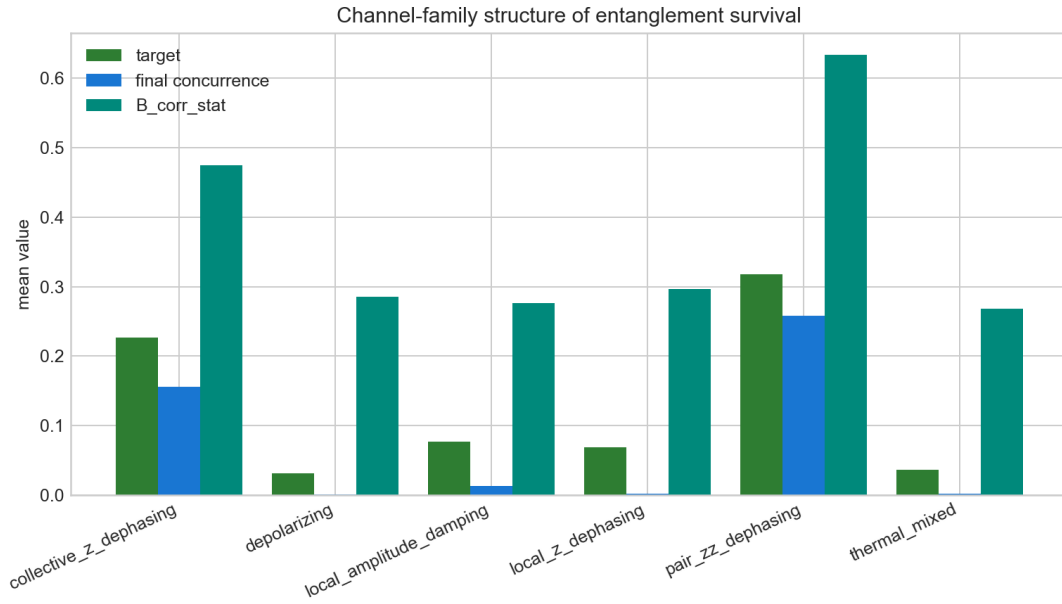


Figure 3: Mean target, final concurrence, and correlated stationarity by channel family. The result is strongly channel-dependent, as expected from decoherence theory.

6.3 Feature Importance

Feature	Permutation importance
R_{rob} stationarity	0.2775
Purity decay pressure	0.1611
Evolution time T	0.1383
Initial concurrence slope	0.0457
Correlated support V_{corr}	0.0350
$B_{\text{corr-stat}}$	0.0275
Correlation drift	0.0257
Exchange coupling J	0.0162
Swap symmetry expectation	0.0144
Depolarizing rate	0.0139

Table 2: Top defect-field feature importances. Stationarity-sensitive robustness, purity decay, and short-time concurrence pressure dominate the learned model.

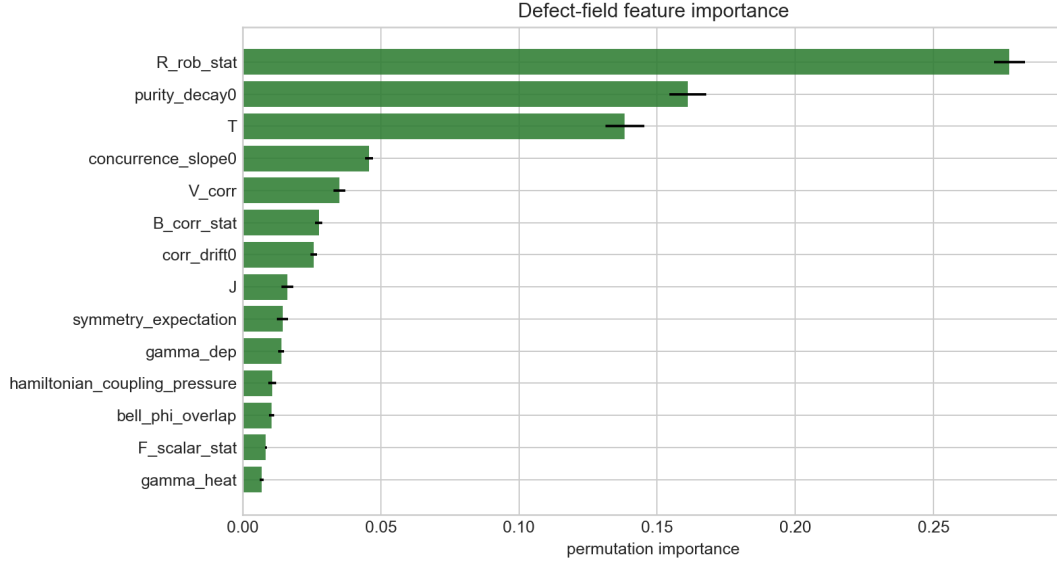


Figure 4: Permutation importance of the full stationary defect-field model.

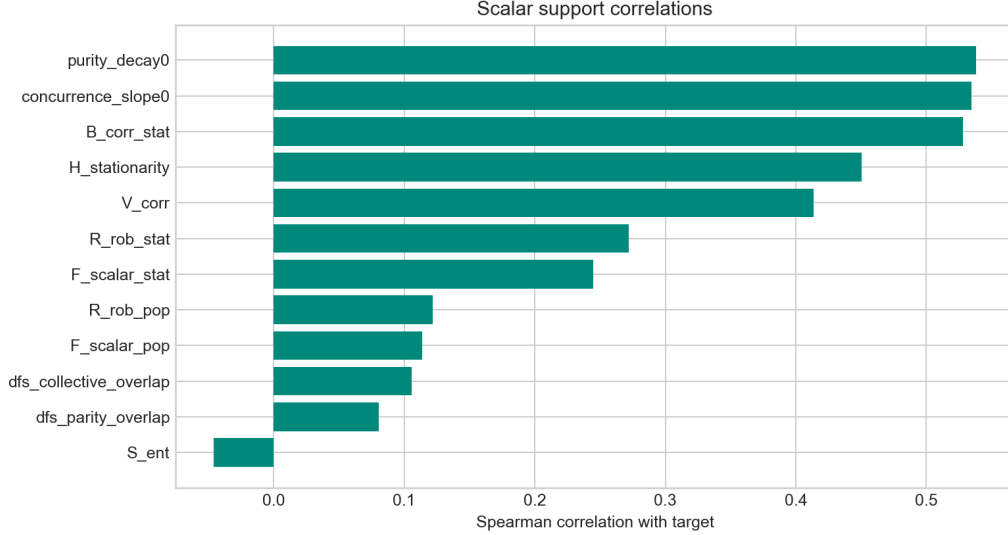


Figure 5: Scalar support correlations with entanglement robustness. The strongest scalar signals are generator-local decay and stationarity pressures, not raw population balance.

7 Interpretation

The result supports the stationarity lesson from Paper 58 in a harder setting. For two qubits, balance cannot mean merely equal local populations. Bell states can be locally balanced and still fragile. What matters is whether the correlated structure is stationary under the specific generator.

The positive result is specific:

correlated generator stationarity improves scalar two-qubit entanglement-robustness prediction over raw population balance.

The stationarity lesson survives, but its practical advantage is currently clearest for in-distribution prediction rather than for out-of-channel generalisation.

The negative result is equally important: the short-time generator baseline remains strongest under leave-channel-family transfer. This means that the current defect-field implementation captures much of the right structure in ordinary cross-validation, but its channel transfer is not yet superior to compact short-time open-system diagnostics.

In decoherence-theory language, the benchmark behaves sensibly: different environments select different robust structures. Collective dephasing and parity dephasing can preserve correlated subspaces, while local damping and depolarization suppress entanglement. MAAT is not replacing this theory. It is providing a defect-field vocabulary for ranking which correlated structures are compatible with a given generator.

8 Limitations

- This is a synthetic classical Lindblad simulation, not quantum hardware.
- The entanglement-robustness target is a transparent benchmark target, not a universal operational definition.
- The channel families are hand-designed and low-dimensional.
- The einselection baseline is lightweight and does not exhaust decoherence theory.

- The full defect-field model wins ordinary cross-validation but not leave-channel-family-out transfer.
- No claim is made about collapse, the Born rule, or the measurement problem.

9 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Experiment folder:

`experiments/two_qubit_pointer_paper64/`

Run:

```
cd experiments/two_qubit_pointer_paper64
python3 two_qubit_pointer_paper64.py
```

Outputs include:

- `paper64_two_qubit_instances.csv`,
- `paper64_model_comparison.csv`,
- `paper64_scalar_correlations.csv`,
- `paper64_feature_importance.csv`,
- `paper64_summary.json`,
- five PNG figures.

Data attribution and license note. No external quantum dataset is redistributed. All trajectories are generated locally from the script using a fixed random seed. CSV, JSON, and PNG files are derived reproducibility artifacts. Repository code is distributed under the MIT License. No endorsement by the authors of Lindblad theory, decoherence theory, or any quantum-computing project is implied.

10 Conclusion

Paper 64 extends the MAAT pointer-state programme from one qubit to two qubits. The central lesson survives: balance should be defined through generator stationarity, not raw population equality.

In this benchmark, correlated stationarity improves scalar prediction of entanglement robustness, and the full stationarity-sensitive defect-field model gives the best ordinary cross-validation performance. But the best leave-channel-family-out result comes from a compact short-time generator baseline.

The mature interpretation is therefore that two-qubit pointer selection is a generator-compatibility problem. MAAT-style defect fields can describe that compatibility, but the next step is stronger cross-channel generalisation rather than larger internal feature sets.

References

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